

Unit 6: Trigonometric Identities and Equations

Precalculus

Student

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6.1 Using Fundamental Identities

Objectives

- Use reciprocal, quotient, Pythagorean, cofunction, and even/odd identities to simplify trigonometric expressions.
- Rewrite trigonometric expressions into equivalent forms.
- Factor trigonometric expressions using algebraic techniques.
- Perform trigonometric substitutions to simplify radical algebraic expressions.

Fundamental Trigonometric Identities

Reciprocal Identities:

$$\sin \theta = \underline{\hspace{2cm}} \quad \cos \theta = \underline{\hspace{2cm}} \quad \tan \theta = \underline{\hspace{2cm}}$$

$$\csc \theta = \underline{\hspace{2cm}} \quad \sec \theta = \underline{\hspace{2cm}} \quad \cot \theta = \underline{\hspace{2cm}}$$

Quotient Identities:

$$\tan \theta = \underline{\hspace{2cm}} \quad \cot \theta = \underline{\hspace{2cm}}$$

Pythagorean Identities:

$$\sin^2 \theta + \cos^2 \theta = \underline{\hspace{2cm}} \quad 1 + \tan^2 \theta = \underline{\hspace{2cm}} \quad 1 + \cot^2 \theta = \underline{\hspace{2cm}}$$

Cofunction Identities:

$$\sin \left(\frac{\pi}{2} - \theta \right) = \underline{\hspace{2cm}} \quad \cos \left(\frac{\pi}{2} - \theta \right) = \underline{\hspace{2cm}}$$

$$\tan \left(\frac{\pi}{2} - \theta \right) = \underline{\hspace{2cm}} \quad \cot \left(\frac{\pi}{2} - \theta \right) = \underline{\hspace{2cm}}$$

$$\sec \left(\frac{\pi}{2} - \theta \right) = \underline{\hspace{2cm}} \quad \csc \left(\frac{\pi}{2} - \theta \right) = \underline{\hspace{2cm}}$$

Even / Odd Identities

An function satisfies $f(-x) = f(x)$, which means its graph is symmetric with respect to the y -axis. An function satisfies $f(-x) = -f(x)$, which means its graph is symmetric with respect to the origin.

Even Identities:

$$\cos(-x) = \underline{\hspace{2cm}} \quad \sec(-x) = \underline{\hspace{2cm}}$$

Odd Identities:

$$\sin(-x) = \underline{\hspace{2cm}} \quad \csc(-x) = \underline{\hspace{2cm}}$$

$$\tan(-x) = \underline{\hspace{2cm}} \quad \cot(-x) = \underline{\hspace{2cm}}$$

Choosing a First Move

Fact

When an expression looks complicated, use the following guidelines to select a starting strategy:

1. **Look for a common factor:** If every term shares a trig function, factor it out first.
2. **Make the terms match:** Use Pythagorean identities to convert squared terms so more parts of the expression use the same trig function.

3. **Treat repeated structure like algebra:** If a trig function appears squared and to the first power, think of it as a quadratic trinomial.
4. **Rewrite reciprocals when they hide cancellation:** Convert functions to sines and cosines to find common denominators or simple cancellations.

Worked Examples

Example 1

Simplify the expression:

$$\sin^2 x \sec x - \sec x$$

Example 2

Rewrite in an equivalent form using only sine:

$$\sin^2 x - \cos^2 x$$

Example 3

Factor the expression:

$$6 \cot^2 \theta - 13 \cot \theta + 5$$

Example 4

Rewrite and factor the expression:

$$\cot^2 x - 2 \csc x - 7$$

Example 5

Simplify the expression:

$$\csc x \sec x - \sin x \sec x$$

Example 6

Rewrite without a fraction:

$$\frac{\sin^2 x}{1 + \cos x}$$

Example 7

Use the substitution $x = 9 \cos \theta$, where $0 < \theta < \frac{\pi}{2}$, to write $\sqrt{81 - x^2}$ as a trigonometric function of θ .

Homework: 6.1 Using Fundamental Identities Homework

6.2 Verifying Trigonometric Identities

Objectives

- Understand the definition of a trigonometric identity.
- Apply the cardinal rule of verification: working on one side of the equation only.
- Use algebraic operations (factoring, finding common denominators, multiplying by conjugates, splitting fractions) and fundamental identities to verify trigonometric identities.
- Compare multiple valid algebraic pathways for verifying the same identity.

What is an Identity?

An _____ is an equation that is true for all values in the domain of the variable. Verifying (or proving) an identity means showing that one side of the equation can be algebraically transformed into the other side using known identities and valid algebra.

The Cardinal Rule: When verifying an identity, you must work on _____ of the equation at a time. You cannot _____ (i.e., do not add, subtract, multiply, divide, or square both sides of the equation). Each step must transform the expression on one side only.

Fact

Guidelines for Verifying Identities:

DO:

- Start with the _____ side.
- Find a _____ to combine fractions.
- _____ out common terms or trinomials.
- Convert all functions to _____.
- Multiply the numerator and denominator by a _____ (e.g., $1 + \cos x$ for $1 - \cos x$).
- Apply _____ to simplify squared terms.

DO NOT:

- _____.
- Multiply _____ sides of the equation by a term.
- Add or subtract a term to _____ sides.
- _____ both sides of the equation.

Worked Examples

Example 1

Verify the identity:

$$\frac{\sec^2 \theta - 1}{\sec^2 \theta} = \sin^2 \theta$$

Example 2

Verify the identity from Example 1 another way (by splitting the fraction):

$$\frac{\sec^2 \theta - 1}{\sec^2 \theta} = \sin^2 \theta$$

Example 3

Verify the identity:

$$\frac{1}{1 - \sin \alpha} + \frac{1}{1 + \sin \alpha} = 2 \sec^2 \alpha$$

Example 4

Verify the identity:

$$\tan x + \cot x = \sec x \csc x$$

Example 5

Verify the identity (hint: work with the right-hand side):

$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

Example 6

Verify the identity:

$$\frac{\cot^2 \theta}{1 + \csc \theta} = \frac{1 - \sin \theta}{\sin \theta}$$

Example 7

Verify the identity:

$$\sin^3 x \cos^4 x = (\cos^4 x - \cos^6 x) \sin x$$

Homework: 6.2 Verifying Trigonometric Identities Homework

6.3 Solving Trigonometric Equations

Objectives

- Use standard algebraic techniques (collecting terms, isolating, factoring) to solve trigonometric equations.
- Solve trigonometric equations of quadratic type.
- Identify and write general solutions vs. solutions over a restricted interval.
- Recognize and eliminate extraneous solutions introduced by squaring.
- Solve trigonometric equations involving multiple angles.
- Solve trigonometric equations using calculators when non-special values are involved.
- Rewrite equations using Amplitude-Phase form to solve.

Basic Techniques

To solve a trigonometric equation, use standard algebraic techniques to _____ involved. Once isolated, use the unit circle to identify the angles that satisfy the equation.

Restricted vs. General Solutions:

- If a domain is given (e.g., $0 \leq \theta < 2\pi$), list only the angles in that _____.
- If no domain is specified, write the general solution by adding _____ for

sine/cosine/secant/cosecant, or _____ for tangent/cotangent, where $n \in \mathbb{Z}$.

Checking Solutions: Always substitute your answers back into the original equation to verify them. This is especially critical when squaring both sides of an equation, which can introduce _____.

Note

The Cancellation Danger (Never Divide by a Variable): Do NOT divide both sides of an equation by a trigonometric expression to cancel it. You may be dividing by _____, which eliminates valid solutions.

- For example, in $2 \cos x \sin x = \sin x$, dividing by $\sin x$ yields $2 \cos x = 1 \implies \cos x = 1/2$. However, you have lost all solutions where $\sin x = 0$!
- **Correct move:** Subtract $\sin x$ to set the equation to zero, then _____ out the common term.

Worked Examples

Example 1

Solve where $0 \leq \theta < 2\pi$:

$$\tan^2 \theta - 3 = 0$$

Example 2

Solve for all real solutions of x :

$$\cot x \cos^2 x = 2 \cot x$$

Example 3

Solve where $0 \leq \theta < 2\pi$:

$$2 \cos^2 \theta - 7 \cos \theta + 3 = 0$$

Example 4

Solve for all real solutions of x :

$$2 \sin^2 x + 3 \cos x - 3 = 0$$

Example 5

Solve in $[0, 2\pi)$:

$$\cos x + 1 = \sin x$$

Example 6

Solve multiple angle equation on $[0, 2\pi)$:

$$3 \tan \left(\frac{x}{2} \right) + 3 = 0$$

Example 7

Solve where $0 \leq \theta < 2\pi$ (calculator required):

$$5 \sin \theta \cos \theta + 4 \cos \theta = 0$$

Amplitude-Phase Form Bridge

Amplitude-Phase Form

Any linear combination of sine and cosine with the same period, $a \sin x + b \cos x$, can be compressed into a single shifted sine wave:

$$a \sin x + b \cos x = R \sin(x + \theta)$$

where:

- The new amplitude is $R = \underline{\hspace{2cm}}$.
- The phase angle θ is the angle corresponding to the coordinates (a, b) , meaning:

$$\cos \theta = \underline{\hspace{2cm}} \quad \text{and} \quad \sin \theta = \underline{\hspace{2cm}} \quad (\theta = \text{atan2}(b, a))$$

Example 8

Rewrite $\sqrt{3} \sin x - \cos x$ in amplitude-phase form.

Example 9

Rewrite the special case $\sin x + \cos x$ in amplitude-phase form.

Example 10

Solve the equation $\sqrt{3} \sin x - \cos x = 1$ on $0 \leq x < 2\pi$.

Homework: 6.3 Solving Trigonometric Equations Homework

6.4 Sum and Difference Formulas

Objectives

- Understand the motivation for sum and difference formulas.
- Memorize and apply sum and difference formulas for sine, cosine, and tangent.

- Find exact values of trigonometric functions for non-standard angles (e.g., 15° , $\frac{\pi}{12}$) by decomposing them into special angles.
- Evaluate expressions of the form $\sin(u \pm v)$ given quadrant information and individual trigonometric ratios.
- Simplify expressions by collapsing expanded sum/difference patterns back into a single trigonometric function.

Why These Identities Matter

Special angles on the unit circle are limited (e.g., $30^\circ, 45^\circ, 60^\circ$). Sum and difference formulas allow us to evaluate exact values for a much wider range of angles (e.g., $15^\circ = 45^\circ - 30^\circ$) and rewrite expressions containing variable shifts into a _____.

Sum and Difference Identities

Sine Formulas:

$$\sin(\alpha \pm \beta) = \underline{\hspace{2cm}}$$

Cosine Formulas:

$$\cos(\alpha \pm \beta) = \underline{\hspace{2cm}}$$

Tangent Formulas:

$$\tan(\alpha \pm \beta) = \underline{\hspace{2cm}}$$

Note

Sign Rule Shortcuts:

- Sine keeps the same _____ inside the product sum.
- Cosine _____ the sign between the two products.
- Tangent keeps the sign on top, but flips it in the _____.

Worked Examples

Example 1

Find the exact value of:

$$\cos 15^\circ$$

Example 2

Find the exact value of:

$$\sin\left(\frac{\pi}{12}\right)$$

Example 3

Find the exact value of:

$$\tan 15^\circ$$

Example Practice

Find the exact value of:

$$\cos 75^\circ$$

Example 4

Given $\sin u = \frac{4}{5}$ where $0 < u < \frac{\pi}{2}$, and $\cos v = -\frac{12}{13}$ where $\frac{\pi}{2} < v < \pi$, find $\sin(u + v)$.

Example 5

Simplify to a single number:

$$\sin 220^\circ \cos 5^\circ + \cos 220^\circ \sin 5^\circ$$

Example 6

Simplify:

$$\cos\left(\theta - \frac{3\pi}{2}\right)$$

Example 7

Simplify:

$$\tan(\theta + 3\pi)$$

*Homework: 6.4 Sum and Difference Identities Homework***6.5 Double-Angle and Half-Angle Formulas****Objectives**

- Apply double-angle formulas for sine, cosine, and tangent to evaluate trigonometric values.
- Solve trigonometric equations using double-angle formulas to convert multiple angle terms into single angle terms.
- Recognize expanded double-angle patterns and collapse them into single trigonometric functions.
- Apply half-angle formulas for sine, cosine, and tangent.
- Correctly perform quadrant analysis to determine the sign ($\pm$) of half-angle evaluations.

Double-Angle Identities**Sine Formula:**

$$\sin(2\alpha) = \underline{\hspace{2cm}}$$

Cosine Formulas (Three Forms):

$$\cos(2\alpha) = \underline{\hspace{2cm}}$$

$$\cos(2\alpha) = \underline{\hspace{2cm}}$$

$$\cos(2\alpha) = \underline{\hspace{2cm}}$$

Tangent Formula:

$$\tan(2\alpha) = \underline{\hspace{2cm}}$$

Half-Angle Identities**Sine Formula:**

$$\sin\left(\frac{\theta}{2}\right) = \underline{\hspace{2cm}}$$

Cosine Formula:

$$\cos\left(\frac{\theta}{2}\right) = \underline{\hspace{2cm}}$$

Tangent Formulas:

$$\tan\left(\frac{\theta}{2}\right) = \underline{\hspace{2cm}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Note

The Half-Angle Sign Rule: The sign ($\pm$) in the half-angle formulas is determined by the quadrant of $\underline{\hspace{2cm}} \underline{\hspace{2cm}} \frac{\theta}{2}$ — NOT the quadrant of θ .

- **Example:** If $\frac{3\pi}{2} < \theta < 2\pi$ (Quadrant IV), dividing by 2 gives $\frac{3\pi}{4} < \frac{\theta}{2} < \pi$ (Quadrant II). Since sine is positive and cosine is negative in QII, $\sin \frac{\theta}{2}$ will use the $\underline{\hspace{2cm}}$ sign and $\cos \frac{\theta}{2}$ will use the $\underline{\hspace{2cm}}$ sign.

Worked Examples

Example 1

Find $\sin(2\theta)$, $\cos(2\theta)$, and $\tan(2\theta)$ given $\cos \theta = \frac{5}{13}$ where $\frac{3\pi}{2} < \theta < 2\pi$.

Example 2

Solve where $0 \leq x < 2\pi$:

$$2 \cos x + \sin(2x) = 0$$

Example 3

Use a double-angle identity to rewrite:

$$\cos^2(5\alpha) - \sin^2(5\alpha)$$

Example Practice

Rewrite as a single trigonometric function:

$$1 - 2\sin^2(3x)$$

Example 4

Find the exact value of:

$$\cos 165^\circ$$

Example 5

Given $\sin x = \frac{2}{5}$ where $0 < x < \frac{\pi}{2}$, find $\cos \frac{x}{2}$.

Example 6

Given $\cos \theta = \frac{5}{13}$ where $\frac{3\pi}{2} < \theta < 2\pi$, find $\sin \frac{\theta}{2}$.